

TELECOM CUSTOMER CHURN ANALYSIS REPORT

BUSINESS ANALYTICS PORTFOLIO PROJECT

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EXECUTIVE SUMMARY

This business analysis investigates customer churn patterns in a US-based **telecom company** using data from **7,043** customers. The study identifies **key churn drivers** and offers **targeted strategies** to improve **customer retention** and lifetime value.

- ◆ Approximately **27%** of customers **churned**, posing a significant **threat** to recurring revenue.
- ◆ **Month-to-month contract** holders are nearly **4x more** likely to churn compared to those on **long-term plans**, highlighting the need for **better contract** incentives.
- ◆ Churn is **highest** among **new** users (0–12 months), signaling gaps in onboarding and early engagement.
- ◆ Customers using **electronic checks** have nearly **double** the churn rate of those using **digital payment** methods, indicating a behavioral segmentation opportunity.
- ◆ Users paying **more than \$70/month** show **increased churn**, revealing potential price sensitivity in high-paying segments.

To address these issues, the following data-backed strategies are **recommended**:

- Incentivize **longer contract** commitments
- Launch targeted **loyalty programs** for **new** customers
- Promote **auto-pay** methods to encourage retention
- **Bundle services** for high-paying users to increase perceived value
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These actions aim to reduce **churn**, enhance **loyalty**, and drive **sustainable business growth** through improved customer lifetime value.



INTRODUCTION

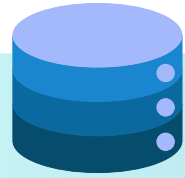
In the competitive landscape of the telecom industry, **customer retention** is not just a metric — it's a lifeline. Subscription-based businesses thrive on loyalty, and even a modest churn rate can result in significant **revenue leakage** and increased **customer acquisition** costs.

This report dives into an **in-depth analysis** of customer churn using a **dataset** from a US-based telecom provider. With over **7,000 customer records**, the objective is to identify the key **behavioral** and **service-related** factors contributing to **churn**, quantify their **impact**, and deliver actionable **strategies to improve** customer lifetime value.

By leveraging **Power BI** for **data visualization** and segment-based **analysis**, this project bridges the gap between raw data and strategic decision-making. The insights drawn aim to equip business leaders with clear, evidence-based direction to **reduce churn**, enhance **customer engagement**, and strengthen competitive positioning in a dynamic market.



METHODOLOGY



DATA SOURCE

- Dataset from Kaggle, titled “Telco Customer Churn”, representing a US-based telecom provider.
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- Total entries: 7,043 customer records
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- Features: 21 columns covering demographics, service usage, billing information, contract types, and churn status.



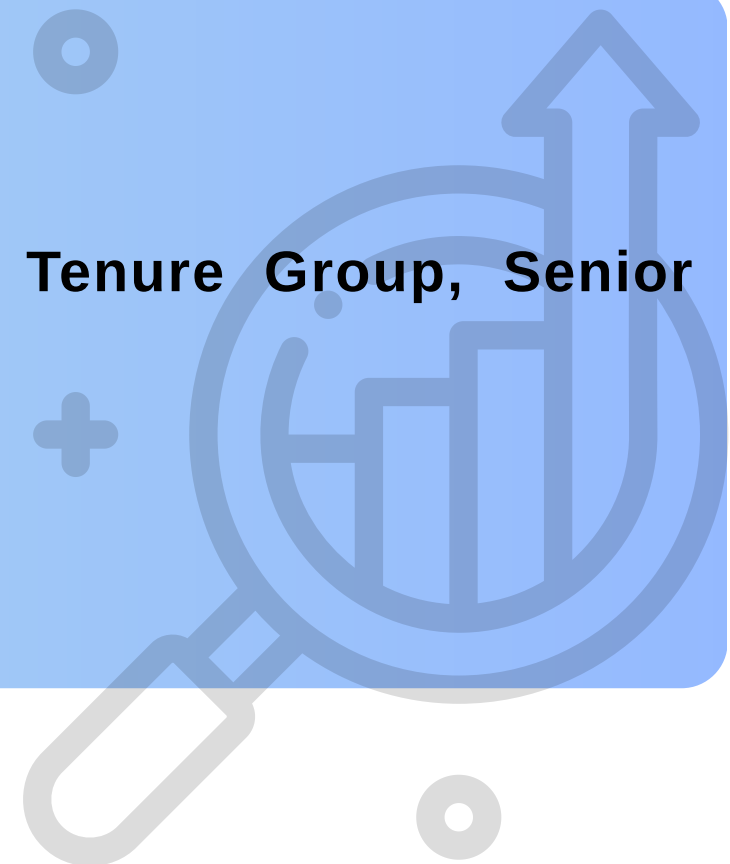
DATA CLEANING & PREPARATION (MICROSOFT EXCEL)

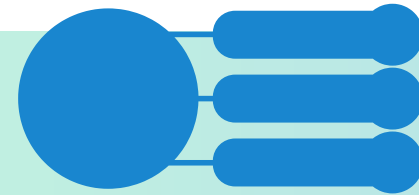
- Blank values: Verified that no missing data was present in critical columns such as TotalCharges.
- Data type corrections:
- Converted TotalCharges from text to numeric.
- Field transformation:
- SeniorCitizen: Recoded 0/1 to “No”/“Yes” for readability.
- Tenure: Binned into groups of 12 months (e.g., 0–12, 13–24...) to analyze lifecycle impact on churn.
- All transformations were made to ensure consistency, interpretability, and compatibility with Power BI.



ANALYTICAL TOOLS USED

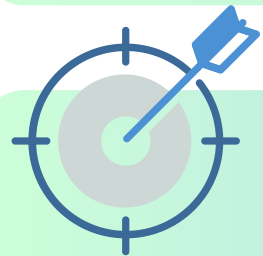
- Power BI (Primary Tool):
- Used for building interactive dashboards and segment-based visuals.
- Allowed dynamic filtering using slicers (Contract Type, Payment Method, Internet Service, Tenure Group, Senior Citizen).
- Created key performance indicators (KPIs) using DAX formulas:
 - $\text{Churn Rate} = \text{COUNT of churned customers} / \text{total customers}$
 - Average Monthly Charges, Total Customers, Average Tenure





VISUALIZATION STRATEGY

- Developed a suite of visuals to extract actionable insights:
- Donut chart to represent churn breakdown
- Bar and column charts to compare churn across service categories (Contract, Tenure, Payment Method)
- Slicers and filters to drill down by customer segments
- KPI cards for high-level summary metrics (churn %, average billing, etc.)



ANALYSIS FOCUS

- Correlated churn status with:
- Contract type (e.g., Month-to-month vs. One-year)
- Customer tenure
- Payment method
- Monthly charge levels
- Patterns were interpreted in the context of behavioral drivers, service friction, and price sensitivity.



SUMMARY

- This structured approach allowed for:
- Clean and consistent data input
 - Clear visualization of relationships
 - Insightful segmentation of churn behavior
 - Business-ready recommendations based on evidence

FINDINGS

1. Churn Rate Is Alarmingly High

- Overall churn rate: ~27%
- Over 1 in 4 customers leave, which significantly impacts recurring revenue
- **Business Implication:** High churn reduces lifetime value and increases customer acquisition costs.

3. Early-Stage Users Are More Likely to Leave

- Churn is highest in the 0–12 month tenure group
- Drops significantly for customers beyond the first year
- **Indicates weak early engagement or onboarding issues**

2. Month-to-Month Contracts Drive Churn

- 43% of month-to-month customers churned
- By contrast, only 11% churned from two-year contracts
- **Customers with flexible contracts are far more likely to leave.**

2. Customers Paying >\$70 Monthly Churn More

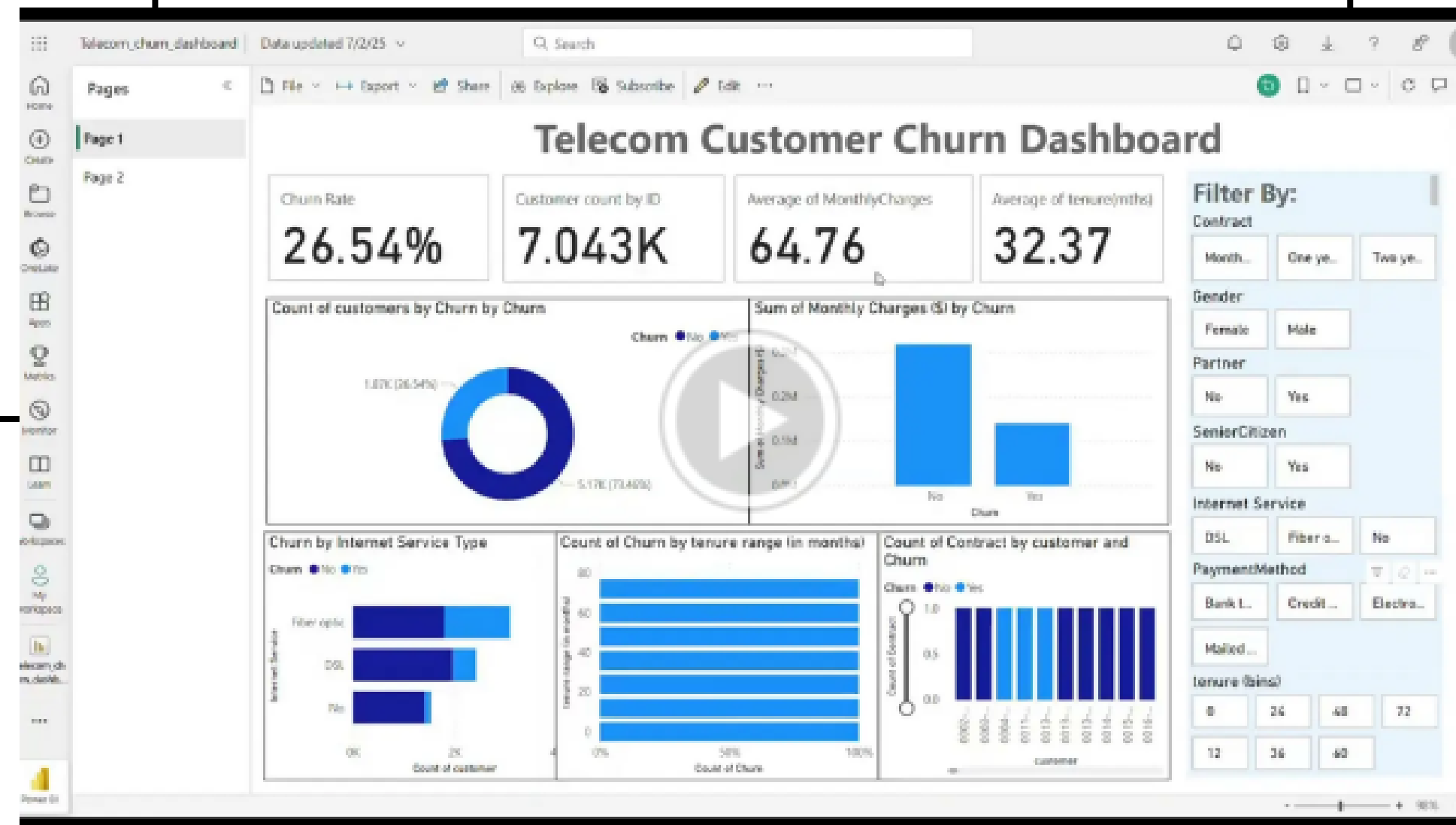
- Users billed above \$70/month show increased churn rates
- Lower-charged users tend to be more stable
- **Suggests price sensitivity — high charges need added value or bundling**

5. Electronic Check Users Churn More

- Customers using electronic checks churn nearly twice as often
- Auto-pay and credit card users show higher retention
- **Digital billing users may represent more engaged or tech-savvy customers**

6. Internet Service Type Impacts Churn

- Fiber optic internet users had higher churn than DSL users
- “No internet service” users had the lowest churn
- **Indicates potential dissatisfaction or unmet expectations with high-speed services**



Power BI dashboard live demo video

RECOMMENDATION

1. Incentivize Long-Term Contracts

- Offer discounts / perks (e.g., extra data, waived activation fees) for switching from month-to-month to 1–2 year contracts
- Highlight cost savings and convenience in marketing

🧠 Targets the 43% churn rate among flexible-plan users

2. Launch First-Year Loyalty Programs

- Provide welcome offers, engagement emails, or retention bonuses during the first 12 months
- Trigger proactive outreach if churn signals appear (e.g., low usage)

🧠 Addresses churn spike in the 0–12 month tenure group

3. Monitor and Improve Fiber Internet Satisfaction

- Conduct feedback surveys for fiber optic users
 - Address service quality concerns or provide alternatives like plan downgrades
 -
- 🧠 Responds to higher churn in fiber internet segment

4. Encourage Digital Payment Adoption

- Promote auto-pay and credit card options with small incentives (e.g., \$5 off first bill, loyalty points)
- Educate users on the security and ease of recurring payments
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🧠 Reduces churn seen in electronic check users



5. Create Value-Added Bundled Plans

- Offer bundled services (e.g., Internet + Phone + Streaming) at a flat or slightly discounted rate
 - Position these plans as premium yet cost-efficient
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- 🧠 Mitigates churn among users billed >\$70/month by improving perceived value

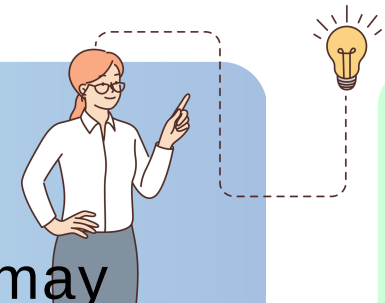
CONCLUSION

Customer churn is a major challenge for telecom providers — our analysis revealed a high 27% churn rate, posing serious risk to recurring revenue and long-term growth.



Through data segmentation, key churn triggers are identified as:

- **Month-to-month contracts (43% churn)**
- **First-year customers (0–12 months)**
- **Manual billing methods (electronic checks)**
- **High monthly charges (>\$70)**



To counter these risks, 6 strategic recommendations may be proposed:

- **Incentivize long-term contracts**
- **Launch loyalty programs for new users**
- **Promote auto-pay and digital billing**
- **Offer bundled service plans**
- **Monitor high-speed service satisfaction**
- **Build real-time churn dashboards**



These actions aim to:

- **Reduce churn by targeting the top behavioral and service-related drivers.**
- **Boost customer lifetime value through longer contracts and bundled offerings.**
- **Enable sustainable growth with focused, data-driven retention strategies.**

THANK YOU !

